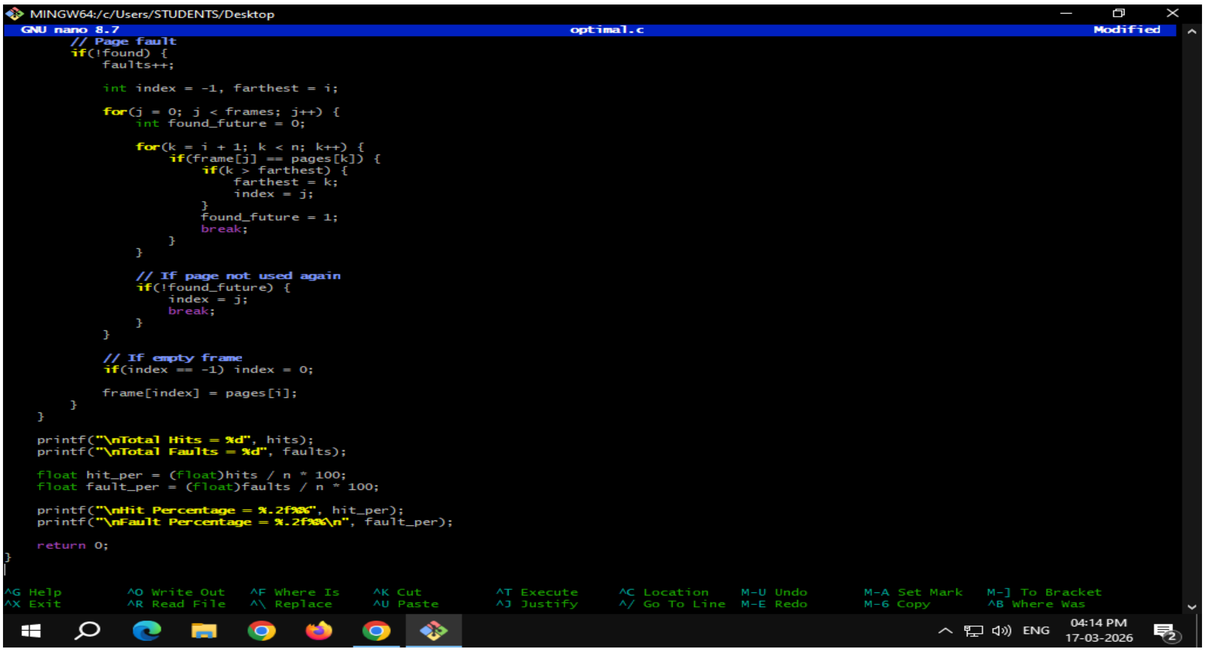
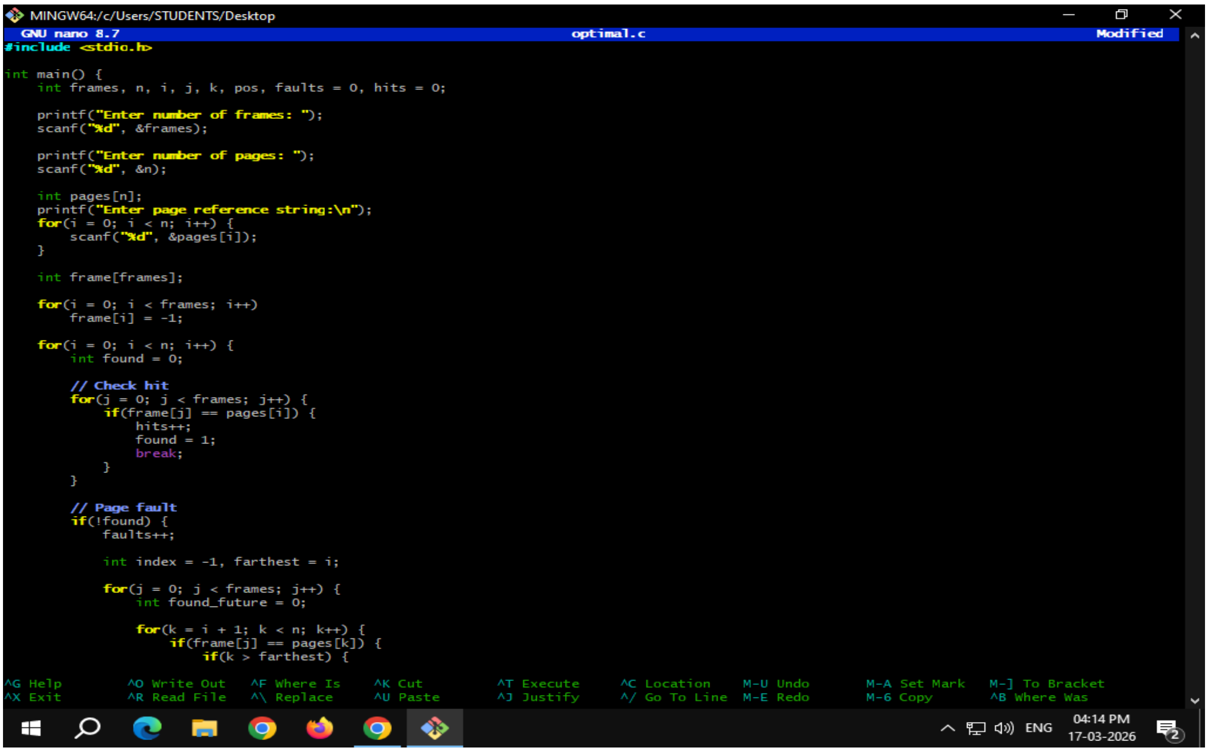
Practical 8

A system has RAM which can store four pages simultaneously to satisfy page reguests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:-

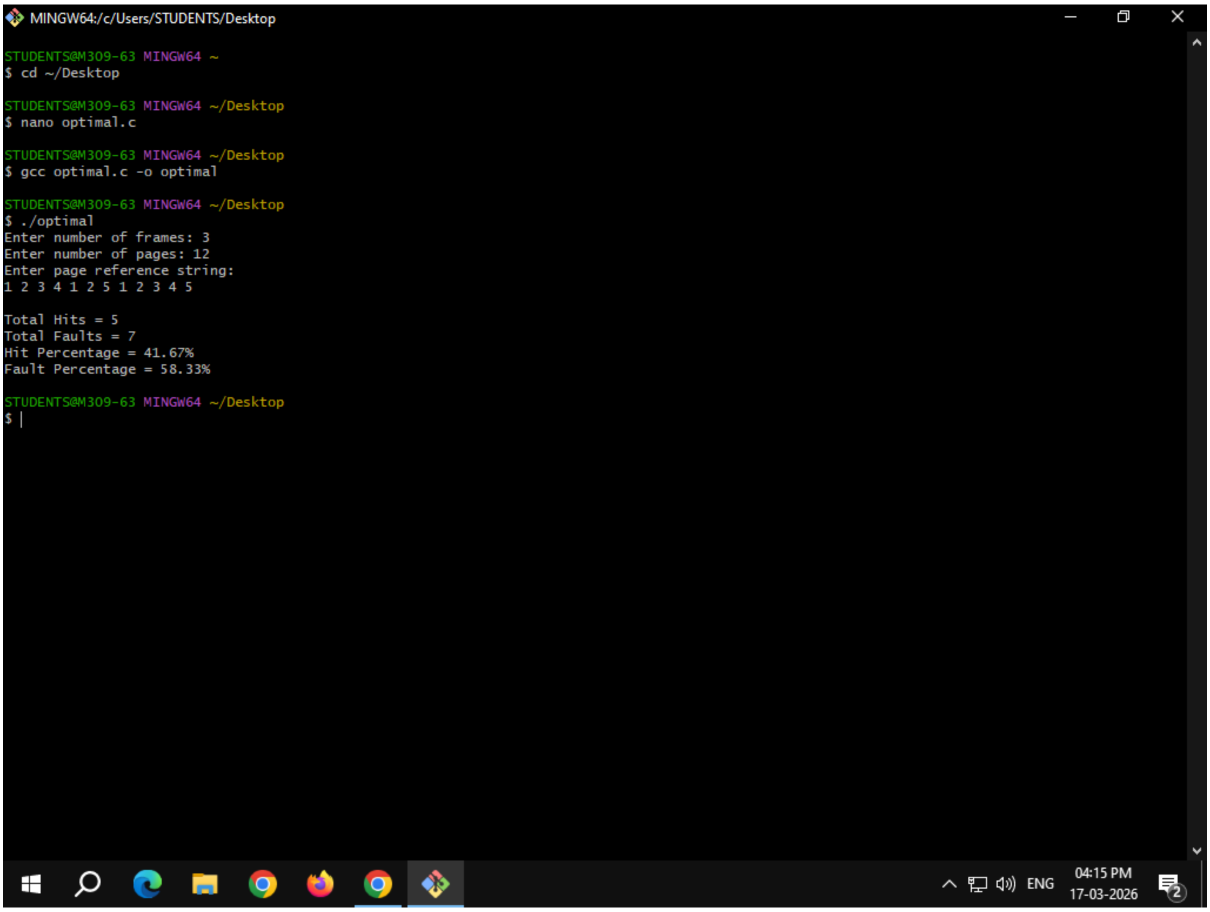


```c
        // Page fault
        if(!found) {
            faults++;

            int index = -1, farthest = i;

            for(j = 0; j < frames; j++) {
                int found_future = 0;

                for(k = i + 1; k < n; k++) {
                    if(frame[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            index = j;
                        }
                        found_future = 1;
                        break;
                    }
                }

                // If page not used again
                if(!found_future) {
                    index = j;
                    break;
                }
            }

            // If empty frame
            if(index == -1) index = 0;

            frame[index] = pages[i];
        }
    }

    printf("\nTotal Hits = %d", hits);
    printf("\nTotal Faults = %d", faults);

    float hit_per = (float)hits / n * 100;
    float fault_per = (float)faults / n * 100;

    printf("\nHit Percentage = %.2f%%", hit_per);
    printf("\nFault Percentage = %.2f%%\n", fault_per);

    return 0;
}
```



```c
#include <stdio.h>

int main() {
    int frames, n, i, j, k, pos, faults = 0, hits = 0;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    int frame[frames];

    for(i = 0; i < frames; i++)
        frame[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check hit
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        // Page fault
        if(!found) {
            faults++;

            int index = -1, farthest = i;

            for(j = 0; j < frames; j++) {
                int found_future = 0;

                for(k = i + 1; k < n; k++) {
                    if(frame[j] == pages[k]) {
                        if(k > farthest) {
```

Output:


MINGW64:/c/Users/STUDENTS/Desktop
STUDENTS@M309-63 MINGW64 ~
$ cd ~/Desktop
STUDENTS@M309-63 MINGW64 ~/Desktop
$ nano optimal.c
STUDENTS@M309-63 MINGW64 ~/Desktop
$ gcc optimal.c -o optimal
STUDENTS@M309-63 MINGW64 ~/Desktop
$ ./optimal
Enter number of frames: 3
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Total Hits = 5
Total Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%
STUDENTS@M309-63 MINGW64 ~/Desktop
$
ENG
04:15 PM
17-03-2026